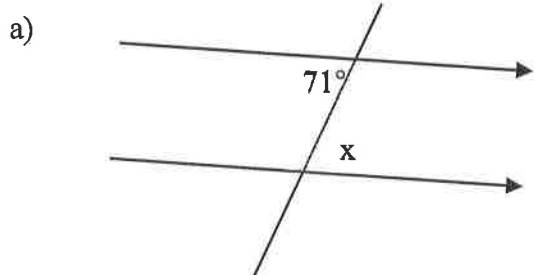


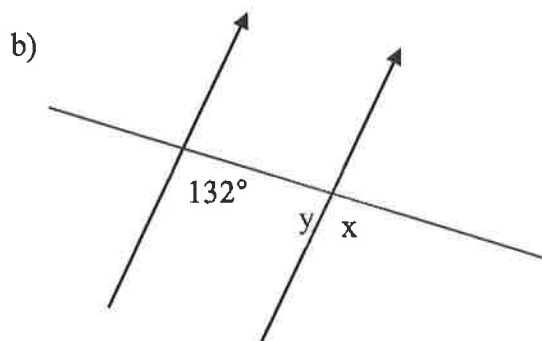
Chapter 7 Intro – Worksheet #3

MPM1D

1. For each problem, determine the value of each unknown and give a reason for your answer.

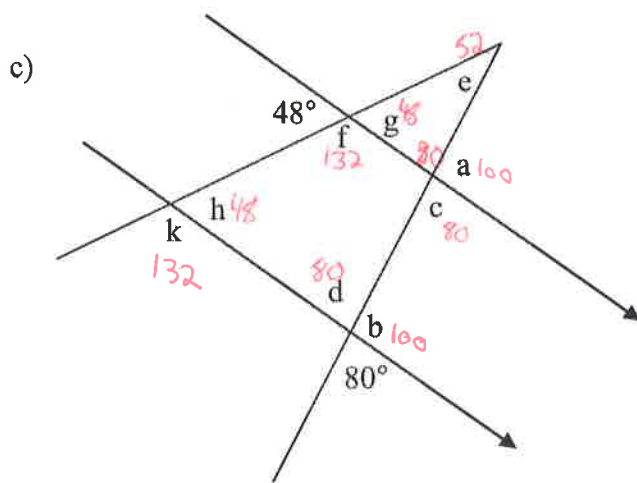


$$x = 71^\circ \text{ (alternate interior)}$$



$$x = 132^\circ \text{ (corresponding)}$$

$$y = 180 - 132 = 48^\circ \text{ (co-interior)}$$



$$d = 80^\circ \text{ (opposite)}$$

$$b = 100^\circ \text{ (supplementary)}$$

$$c = 80^\circ \text{ (alternate int.)}$$

$$a = 100^\circ \text{ (supplementary)}$$

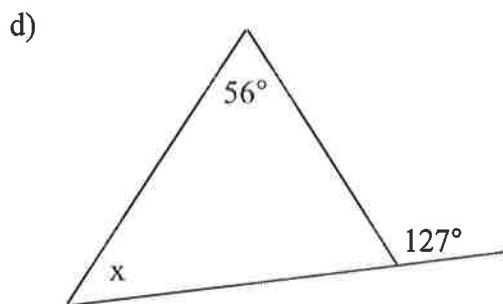
$$f = 132^\circ \text{ (supplementary)}$$

$$g = 48^\circ$$

$$e = 180 - 80 - 48 = 52^\circ \text{ (sum of angles in a triangle)}$$

$$h = 48^\circ \text{ (alternate int.)}$$

$$k = 132^\circ \text{ (supplementary)}$$



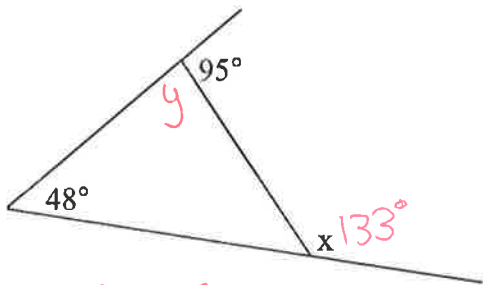
$$56 + x = 127$$

$$x = 127 - 56$$

$$x = 71^\circ$$

(exterior angle)

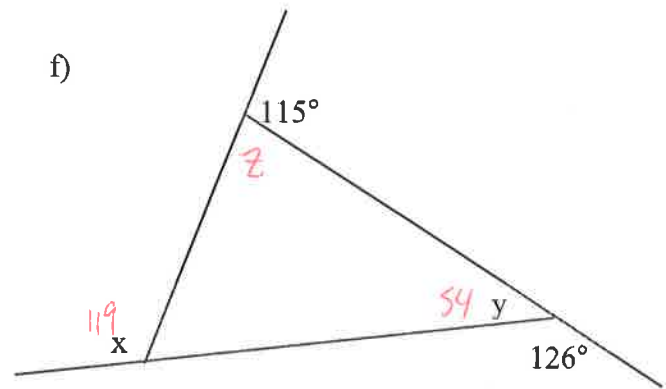
e)



$$y = 180 - 95 = 85^\circ \text{ (supplementary)}$$

$$x = 48 + 85 = 133^\circ \text{ (exterior angle)}$$

f)

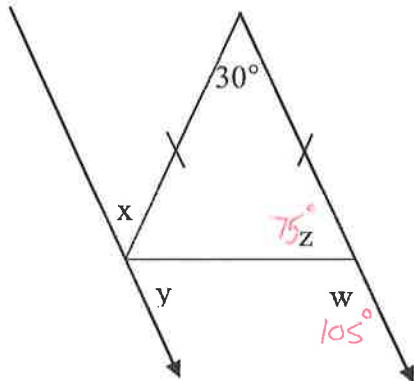


$$y = 180 - 126 = 54^\circ \text{ (supplementary)}$$

$$z = 180 - 115 = 65^\circ \text{ (supplementary)}$$

$$x = 54 + 65 = 119 \text{ (exterior angle)}$$

g)



$$2z + 30 = 180 \text{ (sum of angles in a triangle)}$$

$$2z = 150$$

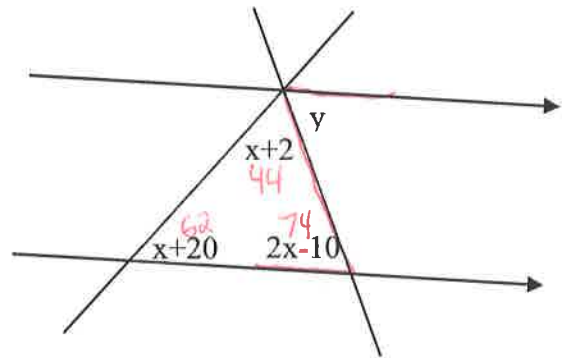
$$z = 75^\circ$$

$$w = 180 - 75 = 105^\circ \text{ (supplementary)}$$

$$y = 180 - 105 = 75^\circ \text{ (co-interior)}$$

$$x = 30^\circ \text{ (alternate interior)}$$

h)



$$x+20 + 2x-10 + x+2 = 180 \text{ (sum of angles)}$$

$$4x + 12 = 180$$

$$4x = 168$$

$$x = 42^\circ$$

$$y = 74^\circ \text{ (alternate interior)}$$

ANSWERS

1) a) $x = 71^\circ$

b) $x = 132^\circ, y = 48^\circ$

c) $a = 100^\circ, b = 100^\circ, c = 80^\circ, d = 80^\circ, e = 52^\circ, f = 132^\circ, g = 48^\circ, h = 48^\circ, k = 132^\circ$

d) $x = 71^\circ$

e) $x = 133^\circ$

f) $x = 119^\circ, y = 54^\circ$

g) $w = 105^\circ, x = 30^\circ, y = 75^\circ, z = 75^\circ$

h) $x = 42, y = 74^\circ$